

Integration by Substitution



u-substitution

1 Evaluate each integral using substitution:

a $\int 2x(x^2 + 1)^5 dx$

b $\int \frac{x}{\sqrt{x^2 + 9}} dx$

c $\int \cos(3x) dx$

2 Evaluate $\int x^2 e^{x^3} dx$.

Definite integrals with substitution

3 Evaluate $\int_0^2 x\sqrt{x^2 + 1} dx$ exactly, changing the limits with $u = x^2 + 1$.

4 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$.

5 Evaluate $\int_1^e \frac{\ln x}{x} dx$.

Substitution with trigonometric and exponential integrands

6 Evaluate each integral using substitution:

a $\int \sin^4 x \cos x dx$

b $\int \frac{e^x}{1 + e^x} dx$

c $\int \sec^2 x \tan x dx$

7 Evaluate $\int \frac{2x + 3}{x^2 + 3x + 1} dx$, recognizing the numerator as the derivative of the denominator.

Choosing the substitution

8 For each integral, state a suitable substitution u and evaluate:

a $\int x^2(x^3 - 1)^7 dx$

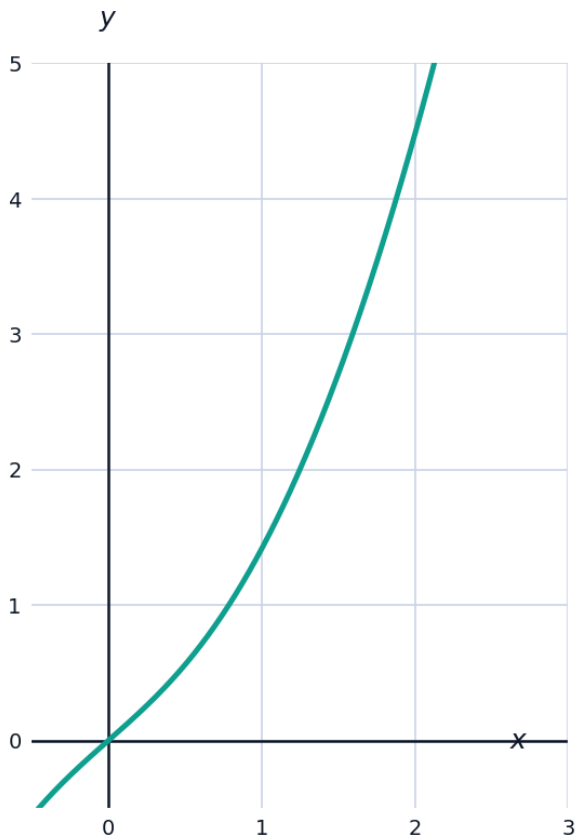
b $\int \tan x dx = \int \frac{\sin x}{\cos x} dx$

Substitution with radicals

- 9 Evaluate $\int x\sqrt{2-x} \, dx$ using the substitution $u = 2 - x$ (so $x = 2 - u$).
- 10 Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}} \, dx$ using $u = x + 1$ (so $x = u - 1$), changing the limits.

Visualizing an area found by substitution

- 11 The graph shows $f(x) = x\sqrt{x^2 + 1}$ on $[0, 2]$. Evaluate $\int_0^2 x\sqrt{x^2 + 1} \, dx$ exactly using $u = x^2 + 1$, and match it to the shaded area.



Substitution with double angle and reciprocal forms

- 12 Evaluate $\int \sin x \cos x \, dx$ two ways: (a) using $u = \sin x$, and (b) using the identity $\sin x \cos x = \frac{1}{2} \sin(2x)$. Confirm both give equivalent answers.
- 13 Evaluate $\int \frac{1}{x \ln x} \, dx$ using $u = \ln x$.

Substitution in motion and rate problems

- 14 A particle's velocity is $v(t) = t\sin(t^2)$ m/s. Find its displacement from $t = 0$ to $t = \sqrt{\pi}$, using $u = t^2$.
- 15 Evaluate $\int \frac{(\ln x)^2}{x} dx$ using $u = \ln x$.
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Recognizing when substitution simplifies a problem

- 16 Evaluate $\int (x + 1)\sqrt{x^2 + 2x + 5} dx$ by noticing the numerator is (up to a constant) the derivative of the expression under the root.